Ning and Yang
Digital Economy and Sustainable Development (2025) 3:24
https://doi.org/10.1007/s44265-025-00071-8

Digital Economy and Sustainable Development

REVIEW

Open Access

A literature review on AI and the global green value chains

Jing Ning[1] and Zheng Yang[2*]

Abstract

Against the backdrop of increasingly severe global climate change and environmental challenges, the traditional energy-intensive and high-emission global value chain model urgently requires transformation towards greener alternatives. The global green value chain (GGVC), by integrating environmentally friendly technologies and practices to optimize resource allocation and reduce carbon emissions, has emerged as a critical pathway for achieving synergistic development between economic growth and environmental protection. Artificial intelligence (AI) technology, leveraging its strengths in data analytics and intelligent decision-making, provides innovative solutions for constructing such green value chains. This study systematically reviews the research progress on the application of AI technology in the GGVC. Through the analysis of core documents, the crucial role of AI in promoting the green transformation of the global value chain has been revealed. The existing literature has found that AI technology is reshaping the development paradigm of the global green economy by empowering innovation throughout the entire industrial chain. In the upstream segments, AI accelerates the research and development of green materials and process optimization. The midstream segments realizes intelligent manufacturing and low-carbon logistics. The downstream links promote green consumption and recycling. This paper constructs a theoretical framework for the development of the global green value chain driven by AI from three dimensions: Technology application, economic impact, and governance challenges, and proposes future research directions.

Keywords Artificial Intelligence, Global Green Value Chains, Sustainable Development, Green Economy

1 Introduction

In recent years, global climate change and environmental degradation have posed severe challenges to sustainable economic development worldwide. The expansion of global trade and deepening international division of labor have closely interconnected environmental issues between developed and developing countries through transnational production activities, bringing international disparities in environmental equity into sharp focus (Bolwig et al. 2010; Golini et al. 2018).The conventional global value chain (GVC) model, characterized by energy-intensive production, high emissions, and low resource efficiency, has not only exacerbated resource and environmental pressures but also facilitated the cross-border transfer of environmental costs (Achabou et al. 2017; Giuliani et al. 2005). This unsustainable development paradigm urgently requires fundamental transformation.

In this context, global green value chain emerges as an innovative development model. The essence of global green value chain is to realize the coordinated development of economic growth and environmental protection by integrating environment-friendly technologies, standards and practices into transnational production and trade activities, optimizing resource allocation, reducing carbon emissions and improving resource recycling

*Correspondence:
Zheng Yang
zhengyang@lnu.edu.cn
[1] School of International Economics and International Relations, Liaoning University, 58 Daoyi South St., Shenbei New District, Shenyang, Liaoning 110136, China
[2] Li Anmin Institute of Economic Research, Liaoning University, 66 Mid-Chongshan Rd, Huanggu District, Shenyang, Liaoning 110036, China

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rate (Gereffi et al. 2016; Schroeder et al. 2018). It is not only related to the sustainable development of individual enterprises, but also an important starting point to achieve the emission reduction target of the Paris Agreement and the United Nations 2030 Agenda for Sustainable Development. It is a key path to promote the sustainable development of the global economy, and provides a new solution to solve the dilemma between economic growth and environmental pressure. However, in the current development process of global green value chain, problems such as data island effect, resource misallocation, regulatory fragmentation and non-uniform standards have become increasingly prominent, which seriously restrict the development of global green value chain (Zhang et al. 2017; Chen et al. 2019).

As a strategic technology leading a new round of scientific and technological revolution and industrial transformation, artificial intelligence has the advantages of self-innovation, cross-border integration and efficient operation, which helps promote green technology innovation, accelerate the transformation of traditional industries to intensive and green, and then lay a solid foundation for accelerating the construction of global green value chain. On the one hand, through the integrated monitoring of automated systems, artificial intelligence can not only accurately predict the trend of energy consumption and carbon emissions, but also simulate the impact of different energy policies and market changes on carbon emissions, helping policymakers formulate scientific and forward-looking climate action plans. It lays a solid foundation for accelerating the green transformation of global value chains (Zhang et al. 2017). On the other hand, artificial intelligence can realize the optimization of production process, time constraint, cost control and other aspects through cloud manufacturing platform and service platform, actively create characteristic industries and functional advantages, further promote the intelligent and green development of traditional industries, and help the green transportation of global value chain (Chen et al. 2019).

Based on this, in-depth exploration of the innovative application and economic impact of artificial intelligence technology in the greening process of global value chain has important theoretical value and practical significance for promoting the sustainable development of global economy. The deep integration of artificial intelligence and global value chain can not only optimize the traditional environmental governance mode (such as carbon emission monitoring and resource allocation), but also promote the green upgrading of all links of the value chain through technological innovation, so as to accelerate the process of sustainable economic and social development, and finally promote the transformation of the whole economic system to a green, low-carbon and efficient direction. This study focuses on the key role of artificial intelligence in the construction of global green value chain, and further discusses the impact of global green value chain based on artificial intelligence technology on the economy, including its role in promoting the green transformation of multinational enterprises, improving the synergy efficiency of global industrial chain, and shaping the new pattern of international green trade. In the long run, the deep integration of artificial intelligence technology is not only the key path to realize the greening of the global value chain, but also has far-reaching strategic significance for formulating scientific low-carbon transformation policies, promoting green technology innovation, and optimizing the pattern of global industrial division of labor.

At present, the research on artificial intelligence in the field of global green value chain mainly focuses on specific aspects, such as supply chain optimization, industrial green transformation or carbon footprint tracking (Patterson et al. 2022; Gebhardt et al. 2022), a comprehensive systematic review of the application of artificial intelligence in the greening of global value chains is still insufficient. To fill this gap, this study provides an extensive review of the application of AI in all links of the global green value chain. This study included a rigorous classification and analysis of 110 AI-related papers and 130 papers dealing with GVCS, green development. In addition, based on existing research, this study divides the global green value chain into three links: upstream, middle and downstream, and conducts an in-depth analysis of each link. Finally, from an economic perspective, this paper discusses how the promotion role of artificial intelligence in the global green value chain can improve the efficiency of relevant enterprises and promote the sustainable development of the global economy.

The rest of the paper is structured as follows (Fig. 1). Section II outlines the research methodology, including manual systematic literature review and paper collection procedures. Section III introduces the theoretical basis and key concepts of artificial intelligence and global green value chain. Section IV comprehensively discusses the application and challenges of artificial intelligence technology in each link of the global green value chain. Section V discusses the economic impact of AI applied to global green value chains at the micro firm and macro economic levels. Section VI discusses the way forward. Section VII concludes the paper.

2 Methodology and data selection

This study adopted the Systematic Literature Review (SLR, hereafter) method proposed by Kitchenham (2007) to collect and analyze relevant articles. SLR is a rigorous

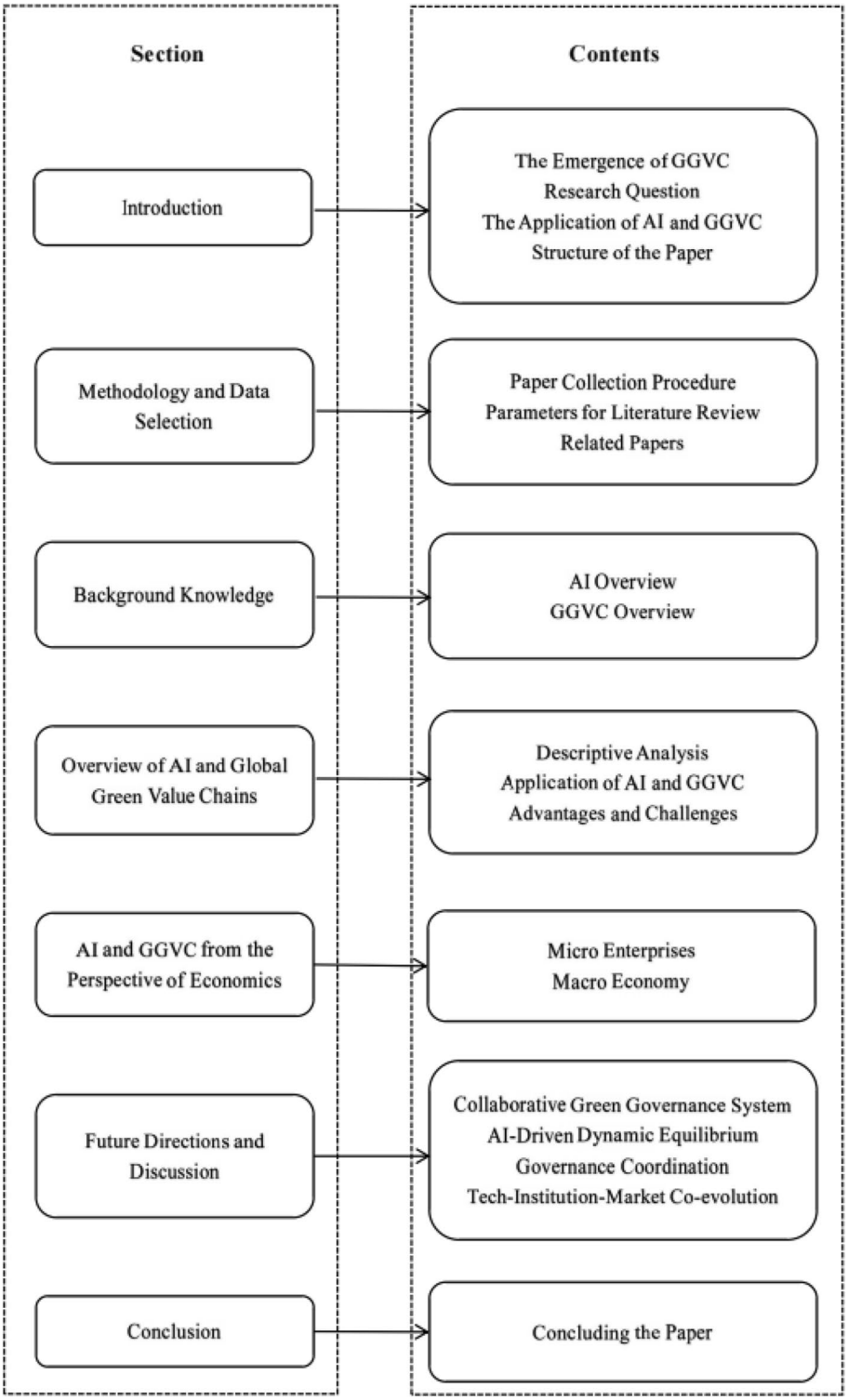

Fig. 1 Structure of the Paper

and structured literature research method, aiming to comprehensively and objectively summarize and analyze the existing achievements in a certain research field through systematic steps. Its workflow can be divided into four key links: First, clearly define the research question and the scope boundary; Secondly, the system

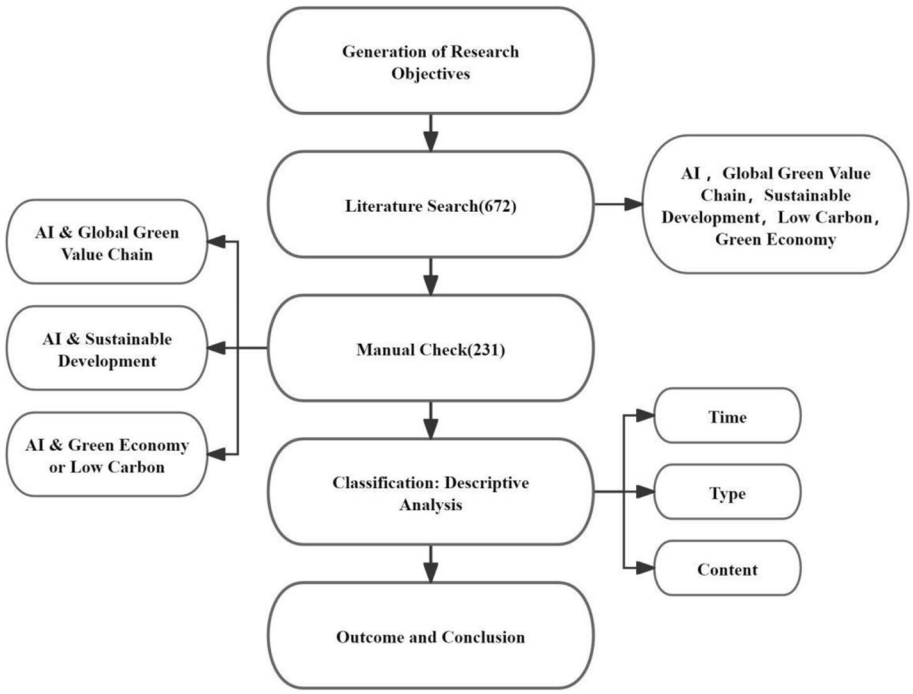

Fig. 2 Literature Selecting and Identifying Procedures

Table 1 Parameters for Literature Review

Parameters	Description
Unit	Articles, Conference Papers
Language	English
Time Range	2005–2025.06
Search String	"Artificial intelligence" AND "Global green value chain" "Artificial intelligence" AND "Sustainable development" "Artificial intelligence" AND "Green economy "OR" Low carbon"
Search Engines	WoS and SCOPUS
Inclusion Criteria	Articles with a focus on Artificial intelligence for global green value chain. English language articles or conference papers of year up to 2005–2025

retrieves core databases such as Web of Science to obtain the initial literature collection; Subsequently, a preliminary screening of the literature is conducted based on the title, abstract, keywords, etc., gradually narrowing down the scope of the literature. Finally, after conducting quality assessment and data extraction on the final selected literature, quantitative (such as bibliometric analysis, meta-analysis) or qualitative (such as thematic analysis, theoretical framework induction) methods were adopted to integrate the research results, thereby systematically answering the research questions. According to this study, the detailed paper collection process is shown in Fig. 2.

Table 1 presents the key parameter settings for the literature search. Since English is the most commonly used international language, English was chosen as the language standard. The time span of this search is from 2005 to June 2025, covering the critical period of the formation and development of global green value chain theory. In view of the significant interdisciplinary characteristics of global green value chain research, and based on the typical application scenarios of AI in environmental governance, we used the combination of three core keywords: "artificial intelligence and global green value chain", "artificial intelligence and sustainable development", and "artificial intelligence and green economy or low carbon"

for retrieval. The search scope is limited to the "title-abstract-keywords" field to ensure the relevance and representativeness of the Literature. This search strategy can effectively capture the cutting-edge research results of AI technology in the fields of green value chain optimization, carbon emission monitoring, and sustainable resource management. In order to ensure the accuracy of Literature research, we systematically screened all search results. By reviewing the title, abstract and keywords one by one, we constructed a complete research framework. After a comprehensive study of the Literature content, 231 high-quality papers[1] that meet the standards of "artificial intelligence and green value chain", "artificial intelligence and sustainable development", "artificial intelligence and green economy or low carbon" are finally screened out.

3 Background knowledge

3.1 Background of AI

3.1.1 Relevant introduction to AI

Artificial intelligence is a new technical science that researches and develops for simulating and expanding human intelligence, which is composed of artificial intelligence theory, method, technology and application system. It is an interdisciplinary subject developed on the basis of the integration of computer, cybernetics, information theory, mathematics, psychology and other disciplines (Li et al. 2017). In terms of the realization process of its algorithmic tasks, AI technology can be classified as a computer engineering technology, and in terms of the realization results and application purposes of its algorithmic tasks, AI technology can be classified as a statistical prediction tool based on modern computer technology.

The development of artificial intelligence can be traced back to the theoretical foundation of artificial neural network in the middle of the twentieth century. In the evolution process of nearly 80 years, artificial intelligence has shown a typical spiral development feature of "technological breakthrough—application landing—bottleneck reflection—breakthrough again" (Lu 2019). This evolution trajectory is not only restricted by internal technical factors such as algorithm innovation and computing power, but also closely related to external factors such as research investment and international economic and political environment.

Artificial intelligence can be divided into three types: narrow AI, general AI and super AI. Narrow AI focuses on specific tasks, such as facial recognition and autonomous driving; General AI aims to match human intelligence and solve complex problems; Super AI can surpass human intelligence and perform better at any task. At present, AI is still in the stage of evolution from narrow AI (ANI) to general AI (AGI), focusing on data mining, intelligent management and decision-making of knowledge and data, and human–computer interaction. With the development of the Internet and the continuous improvement of computer performance, artificial intelligence has made great progress in reinforcement learning, deep learning, machine learning and other fields, forming many research directions such as intelligent robots, language recognition, pattern recognition, image recognition, expert systems, natural language processing and so on, which has become an important part of the current economic and social development.

3.1.2 AI in Economics

Considering that artificial intelligence technology is triggering a new round of scientific and technological revolution and industrial transformation in social production, economists expect to give more interpretation to artificial intelligence technology from the perspective of economics. With the advent of the mobile Internet and big data era, artificial intelligence technology has demonstrated remarkable general technical potential in production applications and has become an important tool for information production and processing. Considering that in the era of Internet big data, data elements have become a new type of production factors, artificial intelligence technology has greatly played the value of data elements and improved the overall economies of scale of social production (Farboodi et al. 2021). From the perspective of the substitution of machines for human labor, the concept of artificial intelligence can be summarized as an automation technology (Acemoglu et al. 2018). The advantage of artificial intelligence technology as an automation technology is that the application of automation technology in production can promote the substitution of capital factors for labor factors, thus having a profound impact on economic growth, employment and inequality through product market and factor market (Jones et al. 2020). (Agrawal, et al., 2019) believed that automation was only the result of the application of artificial intelligence technology, rather than artificial intelligence itself. If artificial intelligence technology is simply defined as automation technology, it seems to largely ignore its intelligent attribute and cannot be distinguished from automation technology since the first industrial revolution. There are still differences in the existing literature on the economic cognition and interpretation of the concept of artificial intelligence.

[1] Out of the 231 papers reviewed, only 100 are cited in the reference list. The remaining papers, while valuable, were not directly referenced as they either overlapped in content or did not align as closely with the specific scope of the study. The full list of reviewed papers is available upon request.

In the economics, artificial intelligence is reshaping the paradigm of traditional economic research and practice by predicting market trends, optimizing resource allocation, simulating policy impacts and improving financial risk management (Ziemba et al. 2025). At the macroeconomic level, deep learning models have been used to improve traditional econometric methods and improve the forecasting accuracy of key indicators such as GDP, inflation rate and unemployment rate (Furman et al. 2019). In microeconomic research, reinforcement learning algorithms help firms optimize pricing strategies, while text analytics techniques are used to mine unstructured data (such as news, social media) to assess market sentiment (Tan et al. 2025; Tero et al. 2025). In the field of economic forecasting, machine learning methods have significantly improved the accuracy of macroeconomic indicator forecasting and policy effect evaluation by processing massive unstructured data (such as social media texts and satellite images) (Goralski et al. 2020). In addition, AI technology itself has also given rise to new economic research fields, including algorithmic economy, data factor market, and AI's impact on the labor market (Zhang 2024).

3.2 Background of global green value chains

3.2.1 The theoretical origin of the global green value chain

The theoretical basis of the concept of global green value chain includes the basic theory of global value chain and green supply chain theory. The theoretical research on global value chain began with the concept of value chain proposed by Porter in his book Competitive Advantage in 1985. The core concepts in the theoretical system of global value chain, such as dual driving mechanism of global value chain, governance mode and upgrading of local industrial cluster, provide basic logical ideas for the research on the greening of global value chain (Humphrey et al. 2002; Gereffi et al. 2005; Gereffi et al. 2012; Porter 1985).

The key link between the green issue and global value chain is supply chain green management theory and its expansion in the global scope. The concept of Green Supply Chain was first proposed by American scholars in a study of Environmentally Responsible Manufacturing (ERM) in 1996. It refers to the modern management mode that comprehensively considers environmental impact and resource efficiency in the whole supply chain. Based on green manufacturing theory and supply chain management technology, it involves suppliers, manufacturers, sellers and users. Its purpose is to make products from material acquisition, processing, packaging, storage, transportation, use to scrap treatment in the whole process. Minimum negative environmental impact and maximum resource efficiency. Since then, driven by enterprises, governments and markets, green supply chain management has been widely accepted by various participants (Webb 1994; Handfield et al. 1996).

Therefore, the global green value chain is a response to the international expansion of green supply chain, and adds issues such as rights, relationships, value distribution and geographical space into green supply chain management (Gereffi et al. 2012; Golini et al. 2016). In general, global value chain is an expansion of the concept of green supply chain in the global scale and value perspective, and the green management of supply chain is also an epitomization of the greenness of global value chain at the company level.

3.2.2 The concept and connotation of global green value chain

According to the existing literature, global green value chain refers to the process in which economic entities change their production methods under the influence of global value chain to reduce environmental hazards, including the organization of green activities and the process of environmental value redistribution in the transnational production process of a series of products from raw material procurement, production, sales, transportation to recycling.It can be divided into two aspects: link greening and collaborative greening, as shown in Fig. 3.

The link greening of GVC is reflected in the green improvement of individual links or processes in GVC (such as procurement, production, recycling, etc.), emphasizing the independent optimization within each link.For example, degradable and recyclable materials are used in raw material procurement, and material reduction is carried out.The collaborative greening of GVC emphasizes the cross-link synergy (technology, circulation and policy linkage) of the overall value chain and the overall issues (such as environmental equity and policy impact), which need to be systematically integrated.For example, to coordinate the relationship between all links of the value chain and realize the green integration from production to recycling through production technology, circulation cooperation and information sharing; Address the distribution of global environmental benefits and environmental equity, and reduce the transfer of polluting industries to developing countries. Analyze the influence of external actors such as national policies on the greening of the value chain.

3.2.3 The driving mechanism of the global green value chain

The driving mechanism of the global green value chain is a dynamic system involving the collaborative efforts of multiple entities, with its core lying in the interaction and checks and balances among international organizations, governments, leading enterprises and consumers.

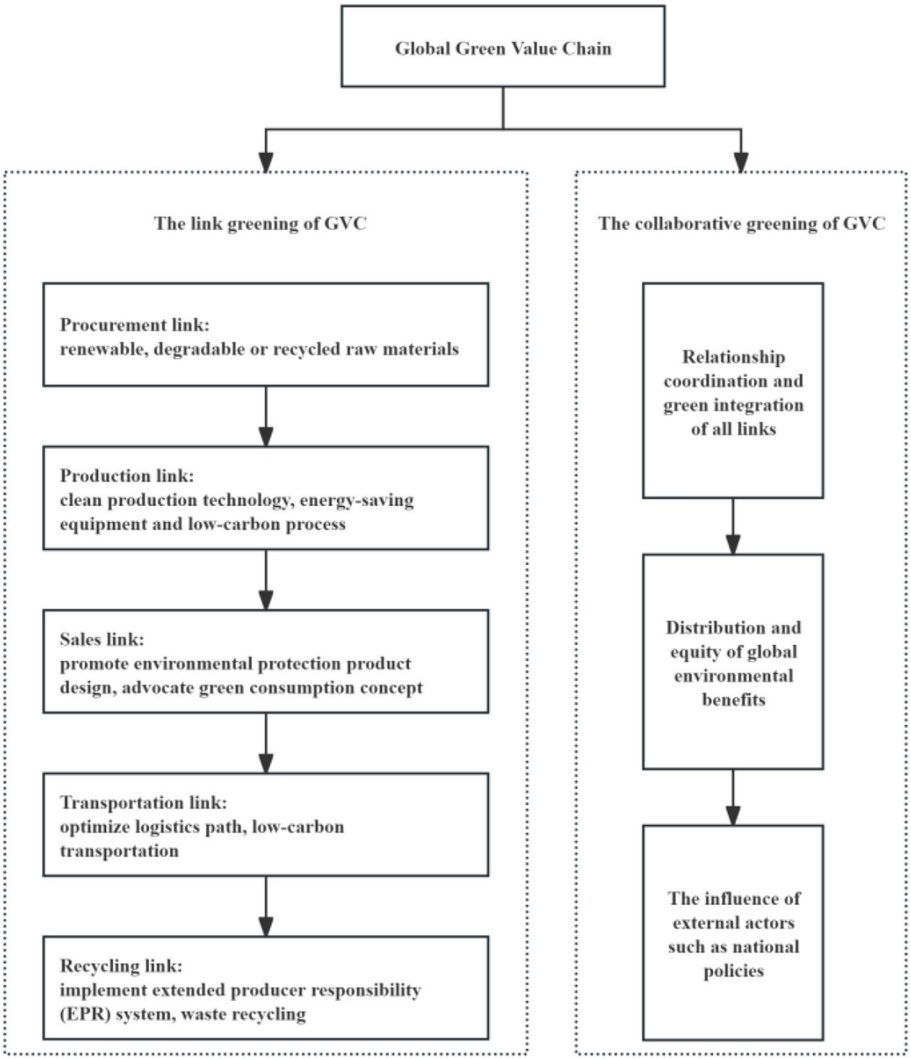

Fig. 3 Conceptual Framework of the Global Green Chain

As shown in Fig. 4, international organizations play a normative role by formulating global environmental standards, supervising compliance and conducting environmental protection publicity, providing an institutional framework for the green value chain. The government relies on "command-and-control" environmental regulations and environmental supervision to force enterprises to internalize environmental costs and guide green practices.Leading enterprises convey environmental protection requirements to downstream enterprises through green supply chain management and promote green technological innovation and diffusion throughout the entire chain by leveraging the spillover effects of knowledge and technology. Meanwhile, consumers' preference for green products and the resistance of environmentally irresponsible enterprises have formed a market-driven mechanism, prompting enterprises to proactively adapt to sustainable demands.

4 Overview of AI and global green value chains

Under the background of the deep integration of global green transformation and digital revolution, artificial intelligence technology is becoming a key enabler to reconstruct the green value chain system. As the goal of carbon neutrality puts forward higher requirements for the transformation of traditional industries, artificial intelligence, through its powerful data analysis and intelligent decision-making ability, can effectively reduce the cost of low-carbon transformation of enterprises and improve the efficiency of resource utilization, which significantly improves the environmental and economic benefits of the global value chain (Vinuesa et al. 2020). The AI-enabled global green value chain is not only the

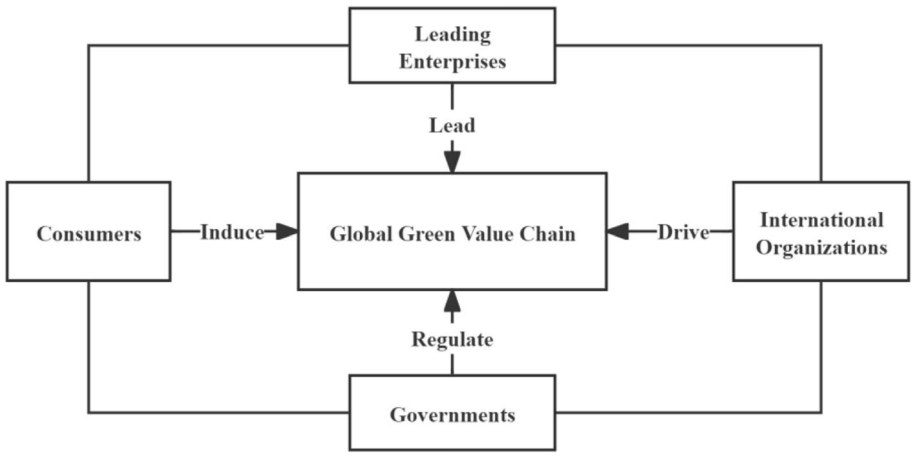

Fig. 4 Driving mechanism of the GGVC

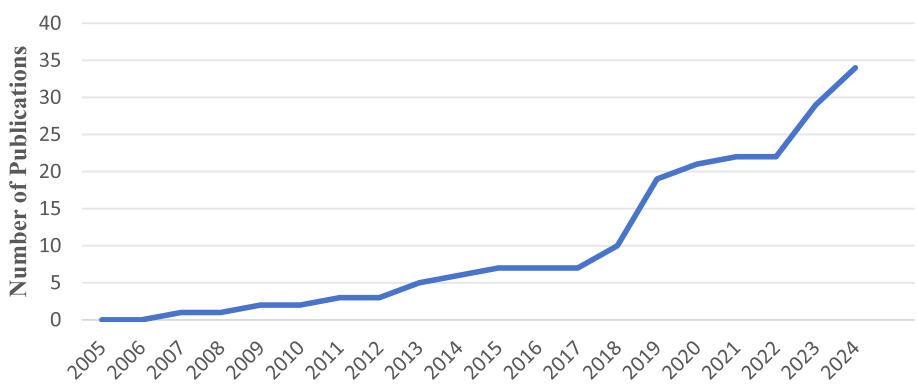

Fig. 5 Year Trend of Publication

key to improving the competitiveness of enterprises, but also an important strategic path for countries to achieve the goal of carbon neutrality. It will become an important pillar of the sustainable development of the global economy in the future, and ultimately promote the transformation of the entire economic system to a green, low-carbon and efficient direction.

4.1 Descriptive analysis

Figure 5 shows that the number of academic publications with AI and global green value chain as core research topics shows a rapid growth trend. This trend is closely related to the rapid development of AI technology and the urgent need for sustainable economic development around the world. Data analysis shows that among the 201 related papers included from 2005 to 2024, approximately 73.1% were published between 2019 and 2024, which reflects that the application of AI technology in the global green value chain is in a period of explosive growth.

Table 2 Type of Articles Related

Type	Number of Articles
Research articles	179
Case study articles	21
Review articles	31
Total	231

Table 2 systematically classifies the types of Literature: 31 are review articles (13.4%), 21 are case studies (9.1%), and 179 are research papers (77.5%). These review articles systematically analyze the application status and development trend of AI technology in carbon footprint tracking, green supply chain optimization, circular economy model innovation and other fields, while exploring the innovative application of machine learning algorithms in environmental compliance supervision, ESG risk assessment and sustainable investment decision-making.

Table 3 Research Contents of the Selected Articles

Research Contents	Number of Articles
Application of AI in the optimization of green value chains	101
Innovation of AI in carbon footprint tracking	77
Ai-driven sustainable economic development	53
Total	231

We thematic classified the research content of 231 papers, which were mainly divided into three categories: the application of AI in green value chain optimization (101 papers), the innovation of AI in carbon footprint tracking (77 papers), and the research on AI-driven economic sustainable development (53 papers). See Table 3 for the detailed classification results, about 77% of the studies focus on the two areas of green value chain optimization and carbon footprint tracking. This classification system fully reflects the multi-cross characteristics of artificial intelligence technology and global green value chain research.

4.2 The application of AI in the global green value chains

In the context of the accelerated transformation of the global green economy, artificial intelligence is becoming the core driving force to promote the upgrading of the whole link of the global green value chain. This paper will systematically analyze the innovative application of artificial intelligence technology in the upstream, midstream and downstream key segment of the global green value chain: the upstream segment focuses on the source innovation of material research and development and process optimization, the midstream segment focuses on the process reform of manufacturing and logistics transportation, and the downstream segment reshapes the terminal transformation of consumption mode and circulation system. These three segments are interconnected and progressive, which together constitute a complete technical map of the global green value chain enabled by artificial intelligence. Through in-depth analysis of typical application scenarios and technical paths of each segment, we can fully grasp how artificial intelligence systematically improves resource efficiency, environmental benefits and economic value of the global green value chain, and contributes to the sustainable development of the global economy.

4.2.1 Upstream segment

The upstream segment is the initial stage of the global green value chain system, covering the whole process from natural resource extraction to primary product processing. The upstream link takes R&D and design as the core, focuses on the source layout of green technology innovation and sustainable solutions, and reduces resource consumption and environmental footprint from the source (Zhao et al. 2024). At this stage, enterprises, scientific research institutions and the government form knowledge sharing networks, build technical barriers and lead the industrial paradigm transformation through the layout of green patents and the formulation of international standards. The innovation achievements of the upstream segment directly determine the green potential of the middle and downstream links, which is the key fulcrum to promote the global value chain to the goal of carbon neutrality, and also the strategic highland for countries to compete for green competitiveness.

In the upstream stage of the global green value chain, AI technology has played a significant role in promoting sustainable development through material innovation and process optimization. In the field of material innovation, the application of artificial intelligence has significantly accelerated the discovery and design process of new environmentally friendly materials. The traditional material research and development model is limited by long development cycle and high trial and error cost, which is difficult to adapt to the rapid development of green manufacturing demand (Jordan et al. 2015). Generative AI provides innovative solutions for materials science research, which greatly improves the development efficiency of green materials through the combination of molecular structure generation algorithms and performance prediction models (Alexopoulos et al. 2020; Orchard et al. 2023). Taking the "generative molecular design" system developed at the Massachusetts Institute of Technology (MIT) as an example, the research team successfully generated chemical structures with specific biodegradability and low toxicity characteristics using the architecture of variational autoencoder (VAE) and graph neural network (GNN). This shortens the screening cycle for new materials from months to weeks with traditional methods, significantly improving the efficiency of green materials development (Li and Zhang 2005; Govindan 2018). In addition, natural language processing (NLP) technology has also been applied to extract material property data from large-scale scientific research literature to build a trainable material database, which also provides important data support for the sustainability assessment of new materials (Zhao et al. 2024).

In the field of process optimization, manufacturing processes with high energy consumption, such as semiconductor manufacturing, metallurgical processing and chemical production, have become a key area of industrial carbon emissions due to their significant energy consumption characteristics. In this context, reinforcement learning (RL) technology is widely used in energy

consumption optimization and emission control of key process. Specifically, the reinforcement learning algorithm achieves significant energy saving and emission reduction effects by dynamically optimizing key variables such as energy consumption parameters and process gas consumption in the manufacturing process (Sakr et al. 2023). Taking the semiconductor manufacturing industry as an example, the AI etching process optimization system developed by Taiwan Semiconductor Manufacturing Company (TSMC) uses RL model to accurately control the gas ratio and time parameters in the plasma etching process, and the energy consumption per unit wafer production is significantly reduced after the implementation of the system (Senoner et al. 2021). In the metallurgical industry, artificial intelligence technology effectively reduces energy waste and significantly improves carbon emission intensity index by optimizing key process parameters such as sintering temperature curve and liquid steel mixing path (Sun et al. 2021). It is worth noting that artificial intelligence technology can also integrate life cycle assessment data to provide scientific basis for the selection of low-carbon solutions in the process design stage (He et al. 2021). These application examples show that AI technology has important emission reduction potential in the field of industrial process optimization.

In addition to making breakthroughs in traditional fields such as material research and development and process optimization, artificial intelligence has also demonstrated great potential in systematic innovation in resource development. The combination of computer vision and unmanned aerial vehicle (UAV) technology is conducive to achieving precise environmental monitoring of mining activities and reducing environmental disturbances; The integration of generative AI and quantum computing is expected to break through the bottleneck of material simulation and accelerate the molecular structure design of new environmentally friendly materials; At the decision support level, multi-agent systems can simulate the impact of different mining schemes on regional ecosystems and are expected to provide visual assessment tools for policy-making (Henry et al. 2025; Wu et al. 2025).These innovative applications can further reshape the upstream technological paradigm, but their large-scale application is still constrained by challenges such as algorithm transparency, computing power infrastructure, and cross-regional collaboration, which require breakthroughs through the collaborative efforts of industry, academia, and research.

The application of artificial intelligence in the upstream stage of the global green value chain has significantly enhanced R&D efficiency and reduced production costs. This not only reconfigures the industrial competition landscape, enabling enterprises with core AI technologies to gain a first-mover advantage and premium pricing power, but also promotes the transformation and upgrading of the global green value chain from resource-dependent to technology-innovative. This technological empowerment not only creates significant direct economic benefits but also intensifies the "Matthew effect" in the field of green technology, exerting a profound impact on the global sustainable development process.

4.2.2 Midstream segment

The midstream segment of the global green value chain is a key transition stage connecting upstream R&D and design with downstream consumer application, covering the whole process of processing, manufacturing and circulation of end products. The midstream segment takes production and logistics as the core, focuses on the collaborative innovation of green manufacturing technology and intelligent supply chain system, and reduces the carbon footprint of products through emission reduction in the whole process. Enterprises, technology suppliers and logistics service providers build industrial cooperation networks at this stage, and reshape the traditional production and circulation mode through the integrated application of clean production and intelligent logistics system. The green transformation effect of the midstream segment directly determines the collaborative emission reduction potential of the upstream and downstream segment, which is the core hub to promote the global value chain to realize the circular economy transformation, and also the key focus of enterprises to practice ESG strategy.

The application of artificial intelligence technology in the middle reaches of the global green value chain is mainly reflected in two aspects: the greening of production process and the low-carbon of logistics and transportation. In the field of production and manufacturing, artificial intelligence promotes the transformation of product production and manufacturing to the direction of precision, intelligence and low-carbon through the integrated application of cutting-edge technologies such as computer vision, digital twin and reinforcement learning (Chen et al. 2024). First of all, computer vision technology driven by deep learning achieves accurate identification of production defects. For example, the visual inspection system deployed by Tesla in its battery manufacturing factory improves the accuracy of assembly defect identification while reducing the product rejection rate (Huang et al. 2021).Secondly, Digital Twin technology, as an emerging intelligent tool, realizes real-time optimization of production scheduling, energy flow and equipment operation status by constructing virtual mapping of production processes (Liu et al.

2021; Tao et al. 2017). The application case of Baosteel Group shows that its AI-enabled digital twin blast furnace model can significantly reduce carbon emissions and improve energy utilization efficiency (Zhang et al. 2023).In addition, reinforcement learning algorithm effectively balances the relationship between energy consumption and capacity output through autonomous learning and dynamic optimization ability in production scheduling, and realizes the intelligent improvement of energy efficiency of manufacturing system (Sakr et al. 2023). For example, the semiconductor industry adopts an intelligent scheduling system based on deep reinforcement learning to dynamically optimize wafer processing sequences and equipment start-stop strategies by analyzing multi-dimensional data such as equipment energy consumption, process parameters and production demand in real time, so as to significantly reduce no-load energy consumption and peak power demand while ensuring stable production capacity (Wang et al. 2023). These technologies are being promoted at an accelerated pace in energy-intensive industries such as chemical and automobile manufacturing, and have become the core technology path for the green transformation of midstream manufacturing.

The application of artificial intelligence technology in logistics is mainly reflected in three key dimensions: path optimization, intelligent scheduling and low-carbon transportation substitution, injecting intelligent and green new momentum into the traditional logistics industry. First of all, artificial intelligence technology is profoundly changing the path optimization mode in the field of logistics and transportation, and realizing the dual improvement of transportation efficiency and green and low-carbon through intelligent algorithms. In the field of shipping, represented by the intelligent navigation system jointly developed by Maersk and Google, it conducts dynamic analysis on a number of real-time parameters such as weather, ocean current and port congestion based on deep reinforcement learning technology, and continuously optimizes ship route planning, which not only improves fuel efficiency, but also greatly reduces invalid voyages through intelligent scheduling. It brings significant carbon emission reduction benefits (Luis and Jorge 2013). In terms of land transportation, logistics enterprises represented by UPS innovatively apply computer vision and machine learning technology to accurately match the cargo volume and transport vehicle space through 3D scanning and intelligent algorithms, which not only improves the utilization rate of container space, but also effectively reduces the transportation energy consumption per unit of cargo (Gebhardt et al. 2022). These innovative practices show that intelligent path optimization technology driven by artificial intelligence has become an important support for modern green logistics system.

Secondly, artificial intelligence technology is promoting the innovation of logistics scheduling system to the direction of intelligence and green. The application of digital twin technology has built a digital mirror of the logistics system. For example, the intelligent scheduling platform connected by 5G network in the Port of Rotterdam has realized real-time processing and dynamic optimization of tens of thousands of data points per second, which not only improves the operation efficiency, but also greatly reduces the operation time of auxiliary engines during the period of ships berthing through precise scheduling. Reduce energy waste (Ivanov and Dolgui 2020). In terms of complex logistics network coordination, multi-agent reinforcement learning technology has shown breakthrough advantages. Taking DHL's "Intelligent Fleet Brain" system as an example, the system uses the Nash equilibrium principle in game theory to achieve efficient collaborative scheduling of thousands of transportation units and achieve carbon emission reduction targets while ensuring service quality (Kar et al. 2022). These innovative practices indicate that artificial intelligence is reshaping the operation mode and sustainable development path of modern logistics system.

Finally, in terms of low-carbon transportation alternatives, represented by the intelligent decision system developed by IBM for P&G, the mixed integer programming and machine learning fusion algorithm is adopted to comprehensively consider multi-dimensional objectives such as carbon emission, cost and timeliness, and the proportion of railway transportation in the European supply chain is successfully increased by 16 percentage points. Achieve remarkable results of reducing emissions by nearly 100,000 tons per year (IBM 2023). What is more noteworthy is that the current low-carbon transportation innovation is showing a trend of cross-technology integration: blockchain builds a trusted carbon footprint traceability system, quantum computing breaks through the bottleneck of computing power in complex optimization problems, and the synergy of these technologies is reshaping the value creation mode of green logistics (Sun et al. 2022; Wang et al. 2023; Nicoletti et al. 2024).

There is still multi-dimensional potential for innovative applications of artificial intelligence in the midstream of the global green value chain. In terms of manufacturing process optimization, a production parameter autonomous design system based on generative AI is expected to break through the limitations of traditional processes. By simulating millions of possible parameter combinations, it can automatically generate the optimal low-carbon production plan; In terms of supply chain carbon footprint management, the carbon accounting platform

integrating blockchain and AI can automatically collect, verify and calculate carbon emission data throughout the entire life cycle of products, providing a reliable foundation for carbon trading (Hirsi et al. 2025; Wu and Wang 2025).Although the technical feasibility of these cutting-edge applications has been initially verified, they still face industrialization obstacles such as data silos, lack of standards, and uncertain investment return cycles in actual promotion. They require the dual drive of policy guidance and market mechanisms to achieve large-scale implementation (Subhadip 2024).

The application of artificial intelligence in the midstream of the global green value chain has significantly enhanced the operational efficiency of enterprises and reduced environmental costs through intelligent production processes and low-carbon logistics systems. Ai-enabled digital twin systems and intelligent dispatching platforms have given rise to new production organization methods, while also giving rise to emerging service formats such as carbon footprint management and green supply chain finance. They are profoundly changing the value creation model and distribution mechanism of global green manufacturing. This technological penetration has not only generated immediate economic returns through cost reduction and efficiency improvement, but also built a lasting competitive advantage through standard setting and model innovation.

4.2.3 Downstream segment

The downstream segment of global green value chain refers to the use, recycling and final disposal stages of products after they are delivered to end consumers, focusing on the value realization of green consumption guidance and resource recycling. This segment takes intelligent consumption scene and cycle system as the core, reshapes the end management of the whole life cycle of products through artificial intelligence technology, constructs low-carbon choice preference at the consumer end, and optimizes resource regeneration efficiency at the recycling end. The downstream segment directly determines the realization degree of green benefits of upstream technology research and development and midstream production and manufacturing. It is the ultimate guarantee to achieve the goal of carbon neutrality in the global value chain, and also the practice carrier to shape a sustainable lifestyle.

Artificial intelligence technology provides key support for the green development of downstream segment by optimizing consumption decisions, improving resource sorting efficiency and enhancing traceability transparency, so that the environmental benefits of the whole life cycle of products can be systematically improved. In terms of low-carbon choice preference construction at the consumer end, artificial intelligence technology is reshaping consumers' green decision-making mode through personalized recommendation system (Ponte et al. 2009). Recommendation algorithm based on deep learning can incorporate sustainability indicators such as product carbon footprint and environmental label into user portrait analysis, and dynamically generate green consumption suggestions in line with personal preferences (Saleem et al. 2022). Platforms such as Amazon and Alibaba have deployed such systems to guide consumers to actively choose environmentally friendly products through visual carbon labels and environmental points reward mechanisms (Zhang et al. 2023). In addition, the product identification system enabled by computer vision technology enables consumers to obtain complete carbon data of the supply chain through mobile scanning, which significantly improves the information transparency of green consumption (Tan et al. 2025).

In the field of intelligent waste recycling, artificial intelligence-driven reverse logistics systems are achieving breakthrough improvements in resource recycling efficiency. Deep learning algorithms predict waste production and distribution characteristics by analyzing historical consumption data, and optimize the dynamic layout of recycling outlets and transportation route planning (Xie et al. 2023). In the sorting process, an intelligent sorting system combining computer vision and robotics can identify and classify wastes of different materials with an accuracy of more than 95% (Wang et al. 2022). The comprehensive application of these technologies builds a closed-loop system from recovery to regeneration, greatly reducing the demand for new material mining.

Blockchain applications are creating new circular business models in product remanufacturing and the sharing economy. The blockchain-based traceability platform records product life cycle data, evaluates residual value and service life combined with AI algorithm, and provides accurate pricing and quality certification for used goods (Oosthuizen et al. 2020). At the same time, the reinforcement learning algorithm optimizes the matching between supply and demand of sharing platforms (Chen et al. 2024). These innovations not only extend the service life of products, but also cultivate a new consumer culture of "use rather than possess".

In the future, artificial intelligence technology is expected to give rise to a number of innovative applications in the downstream links of the global green value chain: Intelligent recommendation systems based on affective computing and consumer behavior analysis will be able to deeply analyze user preferences, generate personalized green consumption plans, and thereby guide the formation of sustainable lifestyles; In the field of

recycling, intelligent sorting centers that integrate computer vision and robotics are expected to achieve automated and precise classification and value assessment of waste. Their recognition accuracy and sorting efficiency will break through the limits of traditional manual operations; In the dimension of business model innovation, AI-enabled sharing economy platforms are expected to significantly enhance product utilization efficiency and effectively curb repetitive production demands by optimizing the intelligent matching algorithm for idle resources (Wei et al. 2022; Sara et al. 2022).Although these forward-looking applications are still in the exploration stage and face practical challenges such as consumer acceptance, data privacy protection, and infrastructure support, the huge potential they demonstrate is expected to reshape the operation mode of downstream links. It is necessary to promote their application by establishing incentive mechanisms and standard systems.

The application of artificial intelligence technology in the downstream links of the global green value chain is triggering profound economic changes. It not only significantly enhances the operational efficiency of each link but also promotes the transformation of the downstream links of the global green value chain from the traditional linear consumption model to a closed-loop economy of "consumption—recycling—regeneration", fundamentally changing the realization path and market distribution mechanism of green value. More importantly, the application of artificial intelligence is giving rise to a series of innovative business models, transforming downstream links from mere terminals of the value chain into new value growth points. The economic impact it brings will continue to deepen and reshape the development pattern of the global green economy.

4.3 Advantages and challenges

4.3.1 Advantages

The application of artificial intelligence in the global green value chain has shown significant advantages and excellent performance. First, by integrating and analyzing massive data across the industrial chain, AI can accurately predict and optimize key links, including raw material procurement, production energy consumption, logistics efficiency and product recycling rate, thus significantly improving the overall resource utilization efficiency. Deep learning algorithm can realize real-time monitoring and intelligent diagnosis of each node of the supply chain, which is helpful to identify environmental risks in advance and optimize decisions. Regardless of the industrial environment, artificial intelligence systems can dynamically adjust operational parameters to ensure that all links of the value chain maintain optimal operation (Kittipanya-ngam et al. 2025; Ta et al. 2024).

Second, AI greatly improves the economic benefits of green value chains. By processing multidimensional data streams, AI technologies are able to optimize various links of the green value chain, including product design, clean production, low-carbon transportation, recycling, and carbon footprint management. Intelligent green management system can reduce the waste of resources in the production process and reduce the cost of environmental compliance. At the same time, artificial intelligence can also ensure the quality stability of green products and reduce the loss caused by supply chain fluctuations, thus significantly improving the market competitiveness of green products and enhancing the sustainable development benefits of enterprises (Tabbakh et al. 2024; Irmina et al. 2024).

Third, AI reshapes strategic decision-making in green value chains. By simulating and predicting complex variables in the green economy, AI provides scientific decision support for enterprises' ESG investment and low-carbon transformation. The real-time analysis generated by machine learning enables companies to dynamically adjust production plans based on regional carbon policies and market demand to achieve precise emission reduction. This helps to formulate an optimal green development strategy. In addition, AI technology enhances supply chain transparency and traceability, ensuring long-term returns on green investments while reducing environmental compliance risks for businesses (Fraga et al. 2021; Megha et al. 2024).

4.3.2 Challenges

As one of the core technologies of the fourth industrial revolution, artificial intelligence has been widely used in green manufacturing, intelligent logistics, carbon footprint tracking and green procurement and other global value chain links. It not only improves the efficiency of resource allocation, but also provides technical support for global carbon neutrality goals. However, despite these significant advantages, the application of AI in the global green value chain still faces multi-dimensional challenges, including technical, economic and policy. These challenges not only restrict the full potential of AI technology, but also affect the overall process of green transformation of global value chains.

AI highly relies on large-scale, high-quality and cross-organizational data in green value chains, but data acquisition faces significant obstacles in the green application of global value chains. First of all, the phenomenon of data islands in each link of the value chain is serious, and the data standards among different enterprises and countries are not unified, which makes it difficult to build a complete carbon emission data chain (Nadvi et al. 2008). Take the apparel industry as an example. The whole value

chain from cotton planting to garment sales involves multiple countries and enterprises, but the energy consumption and emission data of each link often use different measurement standards, making it difficult for AI systems to calculate the carbon footprint accurately. This phenomenon of "data islands" weakens the ability of AI to perform green scheduling and decision support at the system level (Irmina et al. 2024). Second, many developing countries and SMEs have weak digital infrastructure. According to the UNCTAD report, about 70% of SMEs worldwide still use paper record systems, which do not provide sufficient digital data for AI analysis (UNCTAD 2022). Data privacy and trade secrets also pose important barriers. Especially when it comes to suppliers' sensitive data (such as emission data, production capacity and material ratio), enterprises often choose information closure due to competitive strategies. This data closure makes it difficult for the AI system to obtain sufficient training data, affecting the accuracy of the prediction model, and making it difficult to implement the intelligent governance of the green value chain (Lee et al. 2019).

AI technology itself also has many technical challenges in the application of green value chain. Algorithmic transparency comes to the forefront, especially where big business decisions are involved. At present, most deep learning models are black-box operations, and it is difficult to explain their decision logic (Smyth et al. 2022). This lack of explainability makes corporate managers wary of AI-driven environmental decisions. Second, insufficient model generalization ability is another prominent challenge. AI models developed in one industry or region are often difficult to apply directly to other scenarios, such as energy optimization algorithms developed for the European automotive industry, which may be less effective in the Asian textile industry (Robertson et al. 2025). This limitation keeps the cost of scaling AI solutions high, hindering their adoption across the globe. Finally, the difficulty of technology integration cannot be ignored. Embedding AI systems into existing value chain management systems requires complex IT architecture transformation. Most enterprises experience compatibility problems with existing ERP and MES systems when deploying AI sustainable solutions, and these technology integration barriers significantly lengthen project implementation cycles and increase transformation costs (Webber et al. 2019).

The environmental footprint of AI systems raises new concerns. The current development of AI is faced with the "green paradox": although AI can reduce production emissions, its deployment itself causes environmental burdens (Hirsi et al. 2025). Taking large models such as GPT-4 as an example, training a large language model at the GPT-4 level may generate more than 300 tons of CO2 emissions (Drogalis et al. 2023). Without the introduction of "low-carbon AI" strategies, such as green data centers, edge computing, sparse models, etc., it will be difficult to achieve sustainable AI in the true sense (Rožman et al. 2022). Second, algorithmic bias can exacerbate environmental injustice. If training data does not fully reflect different regions and groups, environmental decisions made by AI systems may disproportionately affect some communities. For example, the development based on the data of developed countries leads to poor applicability in the market of developing countries, but increases the overall environmental burden (Wang and Guo 2025). In addition, the risks of technology dependence are increasingly emerging. Over-reliance on AI systems may lead to the degradation of enterprise environmental management capabilities, which may cause serious environmental risks once the system fails or is attacked (Nicoletti et al. 2024).

The lack of a unified policy framework and standard system around the world is an important institutional obstacle to the green application of AI. First of all, carbon emission accounting standards are not uniform. Different countries and industries adopt different measurement methods, which makes it difficult for AI systems to establish a general analysis model (Anderson et al. 2019). A 2023 survey by the International Organization for Standardization (ISO) found that there are more than 20 major carbon accounting standards worldwide, and this fragmentation greatly increases the complexity of developing AI systems (ISO 2023). This inconsistent legal framework limits the unified deployment of AI worldwide and may lead to misuse of the technology and unforeseen environmental impacts (Luis and Jorge 2013). Second, restrictions on the cross-border flow of data hinder AI synergy in global value chains. Many countries place strict restrictions on international sharing of supply chain data due to data sovereignty concerns. For example, the EU's General Data Protection Regulation (GDPR) and China's Data Security Law both put forward strict requirements for data exit. Although these provisions protect privacy and security, they also set up obstacles for building a global AI green value chain management system. Finally, insufficient incentive policies are also a common problem. At present, the environmental protection policies of most countries are still dominated by "command-control", which lacks targeted incentives for the application of AI green technology. According to the World Economic Forum's 2023 report, less than 30 percent of countries around the world have dedicated policies to support the green application of AI, and this absence of policies slows down the rate of technological innovation and diffusion (World Economic Forum 2023).

5 AI and GGVC from the perspective of economics

According to the above research, we find that the impact of artificial intelligence technology on the global green value chain runs through the three links in the upstream, middle and downstream of the industrial chain: the upstream link optimizes resource exploration and green material research and development, the middle link improves the efficiency of clean production and intelligent manufacturing, and the downstream link improves the product recovery and recycling system. Artificial intelligence technology not only systematically improves the green level of the whole industrial chain, but also accelerates the value creation of all links, and promotes the deep transformation of the global economy to low-carbon, circular and digital direction.

5.1 Micro enterprises

The application of artificial intelligence in the global green value chain is profoundly reshaping the operation paradigm and development path of enterprises, and has become a key enabler to promote the green transformation of microeconomy. At present, enterprises are in the strategic opportunity period of green transformation, more and more enterprises have made significant breakthroughs in green technology innovation and operation optimization, and the deep integration of artificial intelligence technology is accelerating this process of transformation. The application of artificial intelligence in the global green value chain is profoundly changing the production mode, cost structure and market competitiveness of enterprises, and bringing significant competitive value and economic benefits to micro enterprises (Ziemba et al. 2025).

In terms of production mode, artificial intelligence technology promotes enterprises to transform from traditional linear production to intelligent circular manufacturing. By integrating industrial Internet of Things and digital twin technology, enterprises have established real-time monitoring and dynamic optimization mechanism of the whole process. For example, enterprises use AI algorithms to deeply analyze massive sensor data of production devices, which significantly improves the utilization efficiency of raw materials and effectively reduces the generation of by-products (Carlos and Casaló 2021). This intelligent transformation not only reshapes the benchmark of resource productivity, but also gives enterprises the ability to quickly respond to the needs of the green market, showing the unique advantages of intelligent manufacturing in green transformation.

In terms of cost optimization, AI reshapes the economic viability of corporate green transformation through a unique value mechanism. In terms of short-term effects, the deployment of AI systems requires enterprises to make considerable upfront capital investment, which is particularly obvious in the construction of intelligent production bases (Davidson and Karell 2025). However, from the perspective of long-term development, the improvement of operational efficiency brought by artificial intelligence can produce continuous cost optimization effect and achieve significant incremental marginal benefits. In particular, it is worth noting that artificial intelligence technology has significantly lowered the application threshold of green innovation, enabling small and medium-sized enterprises to obtain professional carbon management capabilities at a relatively economic cost through cloud-based intelligent service platforms (Fosso et al. 2022). This innovation of cost model effectively broadens the feasible path for enterprises to participate in green economic transformation and promotes a virtuous cycle of scale benefits.

In terms of market competitiveness, AI is redefining the way value is created in green products. First of all, through in-depth analysis of consumer feedback and market trends, AI technology helps enterprises to develop more environmentally attractive product solutions, so that enterprises can gain differentiated advantages in the green market segment (Baabdullah et al. 2021). Secondly, in the field of marketing, the intelligent recommendation system effectively improves the market exposure and purchase conversion rate of environmental protection products, and exposes more consumers to sustainable consumption options (Helo and Hao 2022). In addition, the capital market has also witnessed the changes brought by AI. Intelligent ESG reporting tools not only improve the quality of information disclosure, but also reduce the compliance cost of enterprises. Enterprises that adopt AI technology show obvious advantages in green financing and are more likely to be favored by investors (Li et al. 2021). This AI-driven improvement in market competitiveness enables enterprises that take the lead in layout to take advantage of the wave of green transformation.

Artificial intelligence, by accurately grasping market signals and optimizing decision-making processes, has built an all-round competitive advantage for enterprises, comprehensively empowering their value creation capabilities in the green economy, from product design to marketing, from brand building to capital operation. However, there are still many potential applications and challenges in the process of artificial intelligence promoting enterprises to integrate into the global green value chain. Most enterprises have not yet built an AI-based end-to-end visualization system, which restricts the dynamic optimization ability of the carbon footprint throughout the entire global value chain. Enterprises

generally face a shortage of versatile talents and lack professional teams that are proficient in both green technologies and AI algorithms. Multinational enterprises are particularly constrained by differences in data standards, non-uniform regulatory frameworks, and technological infrastructure gaps, which limit the in-depth application of artificial intelligence in global green supply chain collaboration (Wei and Pardo 2022).This requires enterprises to strengthen the research and development of AI technology and the cultivation of talents, participate in the formulation of international standards and regional collaboration, and build an open and shared industrial ecosystem, so as to fully unleash the full potential of artificial intelligence in the global green value chain.

5.2 Macro economy

The application of artificial intelligence in the global green value chain is profoundly reshaping the operation mode and development trajectory of the world economy and becoming the core driving force for the green transformation of the global economy. At present, the deep integration of artificial intelligence and green value chain is creating a new economic paradigm, and its influence has expanded from micro enterprises to the whole economic system, promoting the fundamental transformation of global production mode and growth mode.

In the field of international trade, artificial intelligence technology is profoundly reshaping the spatial layout and competitive situation of global green value chain. Developed countries, relying on their leading position in artificial intelligence technology, are forming significant advantages in the high-end links of the green value chain. Meanwhile, the role of developing countries in the green value chain is still mainly concentrated in the basic links. This technology gap has also promoted the rise of new modes of international cooperation. The application and promotion of artificial intelligence in the global green field has significantly promoted the growth rate of international trade in green products, making it far faster than the growth rate of traditional commodity trade (Zhang et al. 2025).

Artificial intelligence is accelerating the transformation of traditional industries to green and low-carbon directions, and is a key driving force for global industrial upgrading. The research of the International Energy Agency (IEA) shows that the in-depth application of AI in energy-intensive industries has significantly reduced the carbon emission intensity of global industrial production, especially in traditional energy-intensive industries such as cement, where the production process optimized by AI has brought significant emission reduction results (IEA 2014). What is more noteworthy is that AI technology is cultivating a new green industrial ecology and becoming an important driving force for global economic growth. Taking the "Green digital twin" plan of the European Union as an example, the plan uses AI technology to build a digital model of urban energy system, provides intelligent decision support for the formulation of carbon neutral path, and effectively promotes the process of regional low-carbon transformation (Veugelers et al. 2023). This AI-enabled green innovation model is having a demonstration effect on a global scale.

Artificial intelligence technology is playing a key enabling role in the global green value chain and injecting new impetus into world economic growth. Through intelligent transformation of traditional industrial processes and optimization of resource allocation efficiency, AI technology has significantly improved the overall productivity level of green industries. In the renewable energy sector, AI-driven predictive maintenance and intelligent O&M systems significantly reduce the O&M costs of clean energy equipment; In intelligent manufacturing, process optimization based on machine learning significantly reduces the production energy consumption of green products. These technological innovations not only directly drive the rapid expansion of the scale of green industries, but also create a new growth pole for the global economy by improving total factor productivity (Robertson et al. 2025). At the same time, AI technology is reshaping the pattern of international green trade, enabling developing countries to integrate into the global green economic system faster through technological transition, providing a more sustainable development path for world economic growth (Humphrey et al. 2004). This technology-enabled economic transformation model is promoting the development of global green value chains to be more efficient and inclusive.

Artificial intelligence is accelerating the global green economic transformation and reshaping the industrial, trade and governance landscape. However, AI-driven green transformation may widen the technological gap between developed and developing countries, affecting the fairness and sustainability of the global green transformation. Meanwhile, the fragmentation of the international governance system has also hindered the global spread of green artificial intelligence technologies. Policy differences among various countries and regions in terms of data sovereignty, algorithm transparency, and environmental standards have made it difficult for green AI solutions to achieve economies of scale. Therefore, in the process of promoting the green transformation of the global value chain led by AI, the international community needs to balance innovation and governance, promote technology sharing and rule coordination, and through global cooperation, make AI truly become the

core driving force for the inclusive growth and sustainable development of the global green economy.

6 Future directions and discussion

In the global context of carbon neutrality, the global green value chain is undergoing profound restructuring and multiple challenges. These challenges not only involve breakthroughs in technology, but also require innovation in institutional design, and call for the transformation of global cooperation paradigm. Therefore, the future path of AI-driven global green value chain development will be analyzed below from the aspects of technology integration and innovation, economic and ecological balance, and governance mechanism cooperation, so as to provide ideas for building a more inclusive, efficient and sustainable new order of global green economy.

In the development process of global green value chain, it is difficult for a single technology application to cope with the increasingly complex challenges of sustainable development, so it is urgent to establish a collaborative governance paradigm with deep integration of multiple technologies. Although AI has shown strong potential to optimize decision-making, its effectiveness is still limited by structural bottlenecks such as fragmented supply chain data and difficulties in accounting for carbon footprints. Therefore, integrating digital technologies such as blockchain traceability, satellite remote sensing monitoring and edge computing to build an intelligent environmental management system that runs through the whole product life cycle will become the core engine to promote the transformation and upgrading of the global green value chain (Yu 2022).This technology convergence paradigm can not only track the carbon emission trajectory in real time and accurately quantify the environmental footprint, but more importantly, provide a technical foundation for the establishment of a transparent and credible sustainability assessment system. With the acceleration of global carbon neutrality, this full-chain and multi-dimensional digital environmental governance capability will become a new threshold for enterprises to participate in international competition, and also a key support for countries to achieve climate commitments (Agrawal et al. 2019). In the future, the development of this intelligent management system will reshape the industrial ecology, break green trade barriers through data interconnection, promote the formation of an empirically based carbon pricing mechanism, and finally build a new global green economic order with co-evolution of technology and system.

In the critical period of the transformation of global value chain to green, it is of great significance to explore how artificial intelligence technology can coordinate the dynamic balance of economic benefits and ecological costs in the green value chain. AI not only achieves accurate quantification of environmental costs through smart carbon pricing models, but also optimizes the allocation efficiency of sustainable capital with the help of green fintech, thus reshaping the value-added logic of green industries (Huang et al. 2021).Future research needs to further focus on the mechanism of AI in optimizing the "environment-economy" dual objective function of green value chain, especially on how to use AI technology to promote the leapfrog transfer and inclusive growth of low-carbon technology under the framework of South-South cooperation. At the same time, it is crucial to build an open and collaborative global green AI innovation network, which can match technology supply and demand through intelligent algorithms, accelerate the diffusion of clean technologies, and reduce transaction costs of cross-border green cooperation by relying on blockchain and other digital infrastructure, ultimately forming a systematic solution to reduce green premium and achieve global carbon neutrality. In this process, special attention needs to be paid to AI-enabled transnational data sharing and joint modeling mechanism, so as to overcome the data gap problem of developing countries in green transformation and truly realize the fair reconstruction of global green value chain.

In addition to the analysis of economic benefits, the coordination of digital sovereignty and governance in the global green value chain urgently needs strategic research. With the implementation of new policies such as the carbon Border Adjustment mechanism (CBAM), the complexity of cross-border regulatory compliance is increasing exponentially, which requires us to break away from the traditional policy research paradigm. It is urgent to establish a global policy laboratory based on multi-agent simulation (ABM). A dynamic and interactive carbon market simulation system can be constructed through digital twin technology, which can not only accurately evaluate the differentiated impact of different policy combinations on the industrial competitiveness of each economy, but also accurately evaluate the differential impact of different policy combinations on the industrial competitiveness of each economy. It can better simulate the spillover effect and chain reaction of carbon pricing mechanism between countries in real time (Li et al. 2025). This innovative research method will provide technical support for solving the "prisoner's dilemma" in global green governance, especially in coordinating the burden sharing of carbon emission reduction between developed and developing countries.

In the development process of global green value chain, technology maturity, institutional adaptation and market acceptance form an interdependent three-dimensional driving system. With the full implementation of

mandatory policies such as the Carbon Border Adjustment Mechanism (CBAM) in the European Union, AI-driven green supply chain management has evolved from a competitive advantage to a basic access condition for enterprises to participate in global trade. This changing trend requires the academia to break through the traditional single-dimensional research paradigm, focus on building an evolutionary analysis framework of "technology-institution-market" dynamic coordination, and focus on solving the technological lock-in effect and institutional barriers faced by developing countries in the transition process of green value chain (Agrawal et al. 2021). Therefore, it is urgent to integrate the policy resources of international organizations, the practical experience of multinational companies and the technological capabilities of local innovation centers in an open innovation alliance, and systematically reduce the application cost of green AI technology in small and medium-sized enterprises and emerging economies by developing modular and customizable digital public goods. This collaborative innovation model can not only accelerate the inclusive diffusion of global green technology, but also cultivate market-oriented solutions that adapt to diversified institutional environments, and finally realize the organic unity of technological feasibility, institutional legitimacy and market sustainability in the transformation of green value chain.

7 Conclusion

This paper provides an in-depth discussion of the application of AI technology in the global green value chain and its economic impact. From material research and development and process optimization in the upstream, to intelligent manufacturing and low-carbon logistics in the middle, and to green consumption and recycling in the downstream, artificial intelligence has significantly improved resource utilization efficiency, reduced environmental costs, and reshaped the competitive landscape of the global green economy. We further explore the impact of the application of AI in the global green value chain on economic development from both micro-firm and macro-economy perspectives. Artificial intelligence not only helps micro enterprises reduce environmental costs and enhance green competitiveness, but also promotes macroeconomic growth by fostering new service formats.

Artificial intelligence is becoming a core driver in reshaping the global green value chain. It not only accelerates the process of low-carbon economic transformation, but also provides innovative solutions for the green upgrading of the global industrial chain and promotes the formation of a new value network that is resource-efficient and environment-friendly. With the deepening of technological iteration, artificial intelligence will break through the traditional development bottleneck through the intelligent global coordination mechanism, promote the deep integration of green standards and commercial value, and help build an open, shared, circular and sustainable modern industrial ecosystem. This technology empowerment not only improves the carbon efficiency of all links in the value chain, but also gives rise to a new model of cross-border and cross-field green innovation cooperation, providing systematic digital support for addressing the challenge of climate change.

Authors' contributions
Jing Ning: Original Draft Preparation. Zheng Yang: Supervision, Review and Editing.

Funding
This research received no specific grant from any funding agency.

Declarations

Competing interests
There is no competing interest regarding the publication of this paper.

Received: 9 July 2025 Revised: 13 August 2025 Accepted: 17 August 2025
Published online: 07 October 2025

References

Acemoglu, D., and P. Restrepo. 2018. The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review* 108 (6): 1488–1542.
Achabou, M.A., S. Dekhili, and M. Hamdoun. 2017. Environmental upgrading of developing country firms in global value chains. *Business Strategy and the Environment* 26 (2): 224–238.
Agrawal, A., J. Gans, and A. Goldfarb. 2019. Economic policy for artificial intelligence. *Innovation Policy and the Economy* 19:139–159.
Agrawal, R., V.A. Wankhede, A. Kumar, S. Luthra, and D. Huisingh. 2021. Progress and trends in integrating industry 4.0 within circular economy: A comprehensive literature review and future research propositions. *Business Strategy and the Environment* 31 (1): 559–579.
Anderson, K., M. Bezaitis, and C. DiSalvo. 2019. A.I. among us: agency in a world of cameras and recognition systems. *Ethnographic Praxis in Industry Conference Proceedings* (1): 38–64.
Alexopoulos, K., N. Nikolakis, and G. Chryssolouris. 2020. Digital twin-driven supervised machine learning for the development of artificial intelligence applications inmanufacturing. *International Journal of Computer Integrated Manufacturing* 33 (10): 1–11.
Baabdullah, A.M., A.A. Alalwan, E.L. Slade, R. Raman, and K.F. Khatatneh. 2021. SMEs and artificial intelligence (AI): Antecedents and consequences of AI-based B2B practices. *Industrial Marketing Management* 98: 255–270.
Bolwig, S., S. Ponte, and A. Du Toit. 2010. Integrating poverty and environmental concerns into value chain analysis: A conceptual framework. *Development Policy Review* 28 (2): 173–194.
Chen, W., Y. Men, N. Fuster, C. Osorio, and A.A. Juan. 2024. Artificial intelligence in logistics optimization with sustainable criteria: A review. *Sustainability* 16 (21): 9145.
Chen, Y., Y. Li, and J. Yan. 2019. Tracing air pollutant emissions in China: Structural decomposition and GVC accounting. *Sustainability* 11 (9): 2551.
Carlos, F.,and V.L. Casaló. 2021. Artificial intelligence in services: current trends, benefits and challenges. The Service Industries Journal41 (13-14): 853-859. https://doi.org/10.1080/02642069.2021.1989177

Davidson, T., and D. Karell. 2025. Integrating generative artificial intelligence into social science research: Measurement, prompting, and simulation. *Sociological Methods & Research* 54 (3): 775–793.

Drogalis, M. 2023. How to use GPT-4 with streaming data for real-time generative AI. InfoWorld. com

Farboodi, M., G. Jarosch, and R. Shimer. 2021. Internal and external effects of social distancing in a pandemic. *Journal of Economic Theory* 196: 105293.

Fosso, Wamba S.., M.M. Queiroz, Cameron Guthrie, and Ashley Braganza. 2022. Industry experiences of artificial intelligence (AI): Benefits and challenges in operations and supply chain management. *Production Planning and Control* 33 (16): 1493–1497.

Fraga, P.L., S.I. Lopes, and T.M. Fernández Caramés. 2021. Green IoT and edge AI as key technological enablers for a sustainable digital transition towards a smart circular economy: An Industry 5.0 use case. *Sensors (Basel)* 21 (17): 5745–5745.

Furman, J., and R. Seamans. 2019. AI and the economy. *Innovation Policy and the Economy* 19:161–191.

Gebhardt, M., M. Kopyto, H. Birke, and E. Hartmann. 2022. Industry 4.0 technologies as enablers of collaboration in circular supply chains: A systematic literature review. *International Journal of Production Research* 60 (23): 6967–6995.

Gereffi, G., and J. Lee. 2012. Why the world suddenly cares about global supply chains. *Journal of Supply Chain Management* 48 (3): 24–32.

Gereffi, G., and J. Lee. 2016. Economic and social upgrading in global value chains and industrial clusters: Why governance matters. *Journal of Business Ethics* 133 (1): 25–38.

Gereffi, G., J. Humphrey, and T. Sturgeon. 2005. The governance of global value chains. *Review of International Political Economy* 12 (1): 78–104.

Giuliani, E., C. Pietrobelli, and R. Rabellotti. 2005. Upgrading in global value chains: Lessons from Latin American clusters. *World Development* 33 (4): 549–573.

Golini, R., F. Caniato, and M. Kalchschmidt. 2016. Linking global value chains and supply chain management: Evidence from the electric motors industry. *Production Planning and Control* 27 (11): 934–951.

Golini, R., V. De Marchi, and A. Boffelli. 2018. Which governance structures drive economic, environmental, and social up grading? A quantitative analysis in the assembly industries. *International Journal of Production Economics* 203: 13–23.

Goralski, M.A., and T.K. Tan. 2020. Artificial intelligence and sustainable development. *The International Journal of Management Education* 18 (1): 100330.

Govindan, K., T.C.E. Cheng, N. Mishra, and N. Shukla. 2018. Big data analytics and application for logistics and supply chain management. *Transportation Research Part E: Logistics and Transportation Review* 114: 343–349.

Handfield, R. B. 1996. Green Supply Chain: Best Practices From the Furniture Industry. Proceedings 1996(3): 1295–1297.

He, R., L. Luo, A. Shamsuddin, and Q. Tang. 2021. Corporate carbon accounting: A literature review of carbon accounting research. *Accounting & Finance* 61 (2): 557–581.

Helo, P., and Y. Hao. 2022. Artificial intelligence in operations management and supply chain management: An exploratory case study. *Production Planning & Control* 33 (16): 1573–1590.

Henry, L., and P. Nicolas. 2025. Locating fault for AI harms: A systems theory of foreseeability, reasonable care and causal responsibility in the AI value chain. *Law, Innovation and Technology* 17 (1): 103–138.

Hirsi, A., L. Audah, and A. Salh. 2025. Artificial intelligence performance evaluation for URLLC of industrial IoT applications: A review, open challenges and future directions. *Physical Communication* 72: 102712–102712.

Huang, Z., Y. Shen, J. Li, M. Fey, and C. Brecher. 2021. A survey on AI-driven digital twins in Industry 4.0: Smart manufacturing and advanced robotics. *Sensors (Basel)* 21 (19): 6340.

Humphrey, J., and H. Schmitz. 2002. How does insertion in global value chains affect upgrading in industrial clusters? *Regional Studies* 36 (9): 1017–1027.

Humphrey, J., and H. Schmitz. 2004. Governance in Global value chains. In *Local Enterprises in the Global Economy: Issues of governance and upgrading*, ed. U. K. Cheltenham, 19–29. Edward Elgar.

IBM Research. 2023. Quantum Machine Learning for Sustainable Supply Chain Optimization. *IBM Journal of Research and Development* 67 (3): 1–15.

International Energy Agency (IEA). 2014. Energy technology perspectives 2014. OECD Publishing.

International Organization for Standardization (ISO). 2023. Artificial intelligence-Management system. Retrieved from: https://www.iso.org/standard/81230.html

Irmina, D., T. Miller, E. Kostecka, A. Łobodzińska, and T. Kostecki. 2024. Harnessing AI for sustainable shipping and green ports: Challenges and opportunities. *Applied Sciences* 14 (14): 5994–5994.

Ivanov, D., and A. Dolgui. 2020. Viability of intertwined supply networks: Extending the supply chain resilience angles towards survivability. *International Journal of Production Research* 58 (10): 2904–2915.

Jones, C.I., and C. Tonetti. 2020. Nonrivalry and the economics of data. *American Economic Review* 110 (9): 2819–2858.

Jordan, M.I., and T.M. Mitchell. 2015. Machine learning: Trends, perspectives, and prospects. *Science* 349 (6245): 255–260.

Kar, A. K., S. K. Choudhary, and V. K. Singh. 2022. How can artificial intelligence impact sustainability: A systematic literature review. *Journal of Cleaner Production* 376 : 134120.

Kitchenham, B., and S. Charters. 2007. Guidelines for performing systematic literature reviews in software engineering. EBSE Technical Report 01.

Kittipanya-ngam, P., K.H. Tan, and H.J. Cavite. 2025. Future-ready AI: A framework for ethical and sustainable adoption. *Technology in Society* 83: 102993–102993.

Lee, J., B. Bagheri, and H.A. Kao. 2019. A cyber-physical systems architecture for industry 4.0-based manufacturing systems. *Manufacturing Letters* 3 (1): 18–23.

Li, B., B. Hou, and W. Yu. 2017. Applications of artificial intelligence in intelligent manufacturing: A review. *Frontiers of Information Technology & Electronic Engineering* 18 (1): 86–96.

Li, H.P. 2025. Research on the impact effect of artificial intelligence on the position of China's manufacturing value chain. *Frontiers in Economics and Management* 6 (3): 24–27.

Li, L., and H. Zhang. 2005. Confidentiality and information sharing in supply chain coordination. *SSRN Electronic Journal* 19 : 690862.

Li, M., and T. Li. 2021. AI automation and retailer regret in supply chains. *Production and Operations Management* 31 (1): 83–97.

Liu, M., S. Fang, H. Dong, and C. Xu. 2021. Review of digital twin about concepts, technologies, and industrial applications. *Journal of Manufacturing Systems* 58:346–361.

Luis, A.L., A. Guadalupe, and E.Z. Jorge. 2013. Parcelling virtual carbon in the pollution haven hypothesis. *Energy Economics* 39: 177–186.

Lu, Y. 2019. Artificial intelligence: A survey on evolution, models, applications and future trends. *Journal of Management Analytics* 6 (1): 1–29.

Megha, C., M., and D. R. Sahoo. 2024. Towards a Greener Tomorrow: Exploring the Potential of AI, Blockchain, and IoT in Sustainable Development. Nature Environment and Pollution Technology 23(2):1105–1113.

Nadvi, K. 2008. Global standards, global governance and the organization of global value chains. *Journal of Economic Geography* 8 (3): 323–343.

Nicoletti, B., and A. Appolloni. 2024. Green logistics 5.0: A review of sustainability-oriented innovation with foundation models in logistics. *European Journal of Innovation Management* 27 (9): 542–561.

Oosthuizen, K., E. Botha, J. Robertson, and M. Montecchi. 2020. Artificial intelligence in retail: The AI-enabled value chain. *Australasian Marketing Journal* 29 (2): 264–273.

Orchard, T., and L. Tasiemski. 2023. The rise of generative AI and possible effects on the economy. *Economics and Business Review* 9 (2): 9–26.

Patterson, D., J. Gonzalez, Q. Le, C. Liang, L. Munguia, D. Rothchild, M. Texier, and J. Dean. 2022. Carbon emissions and large neural network training. *Nature Communications* 13 : 1053.

Ponte, S., and J. Ewert. 2009. Which way is "Up" in upgrading? Trajectories of change in the value chain for South African wine. *World Development* 37 (10): 1637–1650.

Porter, M. E. 1985. *Competitive advantage: Creating and sustaining superior performance*. New York, USA: Free Press.

Robertson, J., E. Botha, K. Oosthuizen, and M. Montecchi. 2025. Managing change when integrating artificial intelligence (AI) into the retail value chain: The AI implementation compass. *Journal of Business Research* 189: 115198–115198.

Rožman, M., D. Oreški, and T. Polona. 2022. Integrating artificial intelligence into a talent management model to increase the work engagement and performance of enterprises. *Frontiers in Psychology* 13 (5): 1014434–1014434.

Sakr, A.H., A. Aboelhassan, S. Yacout, and S. Bassetto. 2023. Simulation and deep reinforcement learning for adaptive dispatching in semiconductor manufacturing systems. *Journal of Intelligent Manufacturing* 34 (9): 1311–1324.
Saleem, R., B. Yuan, F. Kurugollu, A. Anjum, and L. Liu. 2022. Explaining deep neural networks: A survey on the global interpretation methods. *Neurocomputing* 513:165–180.
Sara, F., O. Mogren, P. Algurén, and H. Lukas. 2022. Financing solutions for circular business models: Exploring the role of business ecosystems and artificial intelligence. *Business Strategy and the Environment* 32 (6): 3233–3248.
Schroeder, P., P. Dewick, and S. Kusi-Sarpong. 2018. Circular economy and power relations in global value chains: Tensions and trade-offs for lower income countries. *Resources, Conservation and Recycling* 136 (3): 77–78.
Senoner, J., N. Torbjørn, and S. Feuerriegel. 2021. Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science* 83 (6): 19–31.
Smyth, C., S. Gupta, and D. Dennehy. 2022. Artificial intelligence and big data analytics for supply chain resilience: A systematic literature review. *Annals of Operations Research* 327 (2): 21–28.
Subhadip, K. 2024. Data silos - A roadblock for AIOps. *International Journal for Research Trends and Innovation* 9 (1): 2456–3315.
Sun, T., G. Yu, M. Gao, L. Zhao, C. Bai, and W. Yang. 2021. Fault diagnosis methods based on machine learning and its applications for wind turbines: A review. *IEEE Access* 9: 147481–147511.
Sun, X., H. Yu, and W.D. Solvang. 2022. Towards the smart and sustainable transformation of reverse logistics 4.0: A conceptualization and research agenda. *Environmental Science and Pollution Research* 29: 69275–69293.
Ta, P. D. M., S. Wendt, and O. T. Sigurjonsson. 2024. Applying Artificial Intelligence to Promote Sustainability. *Sustainability* 16 (12): 4879–4879.
Tabbakh, A., L. Al Amin, M. Islam, I. Mahmud, I. K. Chowdhury, and M. S. H. Mukta. 2024. Towards sustainable AI: A comprehensive framework for green AI. *Discover Sustainability* 5 (1): 408–408.
Tan, J., S. Chang, and Y. Zheng. 2025. Effects of artificial intelligence in the modern business: Client artificial intelligence application and audit quality. *International Review of Financial Analysis* 104 (9): 104271–104271.
Tao, F., Y. Chen, and L. Zhang. 2017. Digital twin-driven product design, manufacturing and service with big data. *The International Journal of Advanced Manufacturing Technology* 94 (9): 3563–3576.
Tero, S., R. Arto, and K. Osmo. 2025. Manufacturing SME risk management in the era of digitalisation and artificial intelligence: A systematic literature review. *Continuity & Resilience Review* 7 (1): 1–28.
UNCTAD. 2022. Digital economy report 2022: Cross-border data flows and development.
Veugelers, R., C. Faivre, D. Rückert, and C. Weiss. 2023. The green and digital twin transition: EU vs US firms. *Intereconomics* 58 (1): 56–62.
Vinuesa, R., H. Azizpour, I. Leite, M. Balaam, V. Dignum, S. Domisch, and F.F. Nerini. 2020. The role of artificial intelligence in achieving the sustainable development goals. *Nature Communications* 11 (1): 1–10.
Wang, F., Z. Hu, H. Wang, X. Chen, and W. Feng. 2023. Cross-domain dynamic access control based on "blockchain + artificial intelligence." *Neural Computing and Applications* 35: 24575–24585.
Wang, H., and X. Guo. 2025. Forecasting CO_2 emissions in BRICS countries using the grey breakpoint prediction models. *Carbon Balance and Management* 20(1): 7.
Wang, J., W. Ou, W. Wang, R.S. Sherratt, Y. Ren, X. Yu. 2022. Data security storage mechanism based on blockchain network. *Computers, Materials & Continua* 74(3): 4933–4950.
Webb, L. 1994. Green purchasing: Forging a new link in the supply chain. *Resource* 1 (6): 14–18.
Webber, S.S., J. Detjen, and L.T. MacLean. 2019. Team challenges: Is artificial intelligence the solution? *Business Horizons* 62 (6): 741–750.
Wei, R.Q., and C. Pardo. 2022. Artificial intelligence and SMEs: How can B2B SMEs leverage AI platforms to integrate AI technologies? *Industrial Marketing Management* 107: 466–483.
World Economic Forum. 2023. *The Future of AI in Green Value Chains*. Geneva: WEF White Paper.
Wu, J., J. Yao, and Y. Wang. 2025. Platform's role in green vs. non-green product competition: Blockchain transparency and selling contract selection. *International Journal of Production Economics* 288: 109703–109703.
Xie, S., Y. Gong, M. Kunc, Z. Wen, and S. Brown. 2023. The application of blockchain technology in the recycling chain: A state-of-the-art literature review and conceptual framework. *International Journal of Production Research* 61 (24): 8692–8718.
Yu, J. 2022. Enterprise human resource management model by artificial intelligence digital technology. *Computational Intelligence and Neuroscience* 12 (6): 6186811–6186811.
Zhang, J. 2024. Research on the integration path of artificial intelligence and real economy in the digital economy era. *Artificial Intelligence Technology Research* 2 (4): 14–19.
Zhang, L. 2023. Third-generation neural networks for environmental parameter analysis. *Artificial Intelligence Review* 56 (8): 1023–1045.
Zhang, N., W. Liu, and H.C. Wu. 2025. Introduction to the topic of AI and IoT for promoting green operation and sustainable environment. *Systems* 13 (1): 62.
Zhang, Z., K. Zhu, and J. D. Geoffrey. 2017. A multi-regional input-output analysis of the pollution haven hypothesis from the perspective of global production fragmentation. *Energy Economics* 64:13–23.
Zhao, P., Y. Song, Z. Hu, Z. Zhong, Y. Li, K. Zhou, T. Qin, Y. Yan, H.-H. Hsu, S.-T. Han, V.A.L. Roy, C.-C. Kuo, and Y. Zhou. 2024. Artificial intelligence enabled biodegradable all-textile sensor for smart monitoring and recognition. *Nano Energy* 130: 110–118.
Ziemba, W.E., W. Grzenda, and M. Ramsza. 2025. Explainable artificial intelligence for sustainable development: Advancing social and economic transformations. *Artificial Intelligence Review* 67 (8): 892–984.

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